

# BSDS (2025 Semester II): Statistics II

## Midsem: Part B

NAME :

ROLL :

**Time:** 60 minutes

**Total marks:** 26

*Answer all questions.*

1. Let  $X_1, \dots, X_n$  be  $n$  i.i.d. samples from the population  $F_\theta$  with mean  $\mu_\theta$  and finite variance  $\sigma_\theta^2$ , and let  $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$ . Show that  $E(\bar{X}_n) = \mu_\theta$  and  $\text{var}(\bar{X}_n) = n^{-1} \sigma_\theta^2$ . [2 + 3]

$$E(\bar{X}_n) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu_\theta = \mu_\theta. \quad [\text{Ans 1}]$$

$$\begin{aligned} \text{var}(\bar{X}_n) &= \text{var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{var}(X_i) + \frac{1}{n^2} \sum_{i \neq j} \underbrace{\text{cov}(X_i, X_j)}_{0 \text{ as } X_i \perp X_j} \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma_\theta^2 = \frac{\sigma_\theta^2}{n} \quad [\text{Ans 2}] \end{aligned}$$

2. let  $X_1, \dots, X_n$  be a random sample from the log-normal distribution with parameters  $\theta = (\mu, \sigma^2)$  and p.d.f.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma x}} \exp \left\{ -\frac{1}{2\sigma^2} (\log x - \mu)^2 \right\}, \quad \text{for } x > 0, \mu \in \mathbb{R}, \sigma^2 > 0.$$

Find the MLE of  $\theta$ .

[8]

The joint pdf of  $X_1, \dots, X_n$  is:

$$f_{\theta}(x) = \left(\frac{1}{2\pi}\right)^{n/2} \sigma^{-n/2} \prod_{i=1}^n x_i^{-1} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (\log x_i - \mu)^2 \right\} \quad (1)$$

$x_i > 0, \mu \in \mathbb{R}, \sigma^2 > 0$

The log-likelihood is

$$\begin{aligned} l(\theta; x) = & -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \sum_{i=1}^n \log x_i \\ & - \frac{1}{2\sigma^2} \sum_{i=1}^n (\log x_i)^2 + \frac{\mu}{\sigma^2} \sum_{i=1}^n \log x_i - \frac{n\mu^2}{2\sigma^2} \end{aligned}$$

$$\frac{\partial l}{\partial \mu} = 0 \Rightarrow \frac{1}{\sigma^2} \sum_{i=1}^n \log x_i - \frac{n\mu}{\sigma^2} = 0 \Rightarrow \hat{\mu} = \frac{1}{n} \sum_{i=1}^n \log x_i$$

$$\frac{\partial l}{\partial \sigma^2} = 0 \Rightarrow -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (\log x_i - \mu)^2 = 0$$

$$\Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (\log x_i - \hat{\mu})^2$$

Further,

$$\left. \frac{\partial^2 l}{\partial \mu^2} \right|_{\mu=\hat{\mu}; \sigma^2=\hat{\sigma}^2} = -\frac{n}{\hat{\sigma}^2}$$

$$\left. \frac{\partial^2 l}{\partial \sigma^2 \partial \mu} \right|_{\mu=\hat{\mu}; \sigma^2=\hat{\sigma}^2} = -\frac{1}{\hat{\sigma}^4} \sum_{i=1}^n \log x_i + \frac{n\hat{\mu}}{\hat{\sigma}^4} = 0$$

$$\left. \frac{\partial^2 l}{\partial (\sigma^2)^2} \right|_{\sigma^2=\hat{\sigma}^2; \mu=\hat{\mu}} = \frac{n}{2\hat{\sigma}^4} - \frac{1}{\hat{\sigma}^6} \sum_{i=1}^n (\log x_i - \hat{\mu})^2$$

$$= \frac{n}{2\hat{\sigma}^4} - \frac{n\hat{\sigma}^2}{\hat{\sigma}^6} = -\frac{n}{2\hat{\sigma}^4}$$

Hence the Hessian at  $(\hat{\mu}, \hat{\sigma}^2)$  is:

$$H(\hat{\mu}, \hat{\sigma}^2) = \begin{bmatrix} -\frac{n}{\hat{\sigma}^2} & 0 \\ 0 & -\frac{n}{2\hat{\sigma}^4} \end{bmatrix} ; \text{ which is negative definite.}$$

Hence the MLE is :

$$\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n \log x_i$$

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (\log x_i - \hat{\mu})^2$$

$$=$$

3. Find the UMVUE of the parameters  $\mu$  and  $\sigma^2$  for log-normal distribution (described in Q.2). [8]

From the joint-pdf of  $X_1, \dots, X_n$  in (1) we get:

$$f_{\theta}(\underline{x}) = \exp \left\{ -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \sum_{i=1}^n \log x_i - \frac{1}{2\sigma^2} \sum_{i=1}^n (\log x_i)^2 + \frac{1}{\sigma^2} \mu \sum_{i=1}^n \log x_i - \frac{n\mu^2}{2\sigma^2} \right\}$$

Then  $\left\{ \sum_{i=1}^n (\log x_i)^2, \sum_{i=1}^n \log x_i \right\}$  is the Complete Sufficient statistics (CSS) for  $(\mu, \sigma^2)$ .

We will first find the expectation of  $\log x_1$  and that of  $(\log x_1)^2$ .

Define  $y = \log x_1$ , i.e.,  $x_1 = e^y$

Then  $\frac{dy}{dx_1} = \frac{1}{x_1} = e^{-y}$ , range 

|       |           |          |
|-------|-----------|----------|
| $x_1$ | 0         | $\infty$ |
| $y$   | $-\infty$ | $\infty$ |

,

and Jacobian  $e^y$ .

$$\begin{aligned} f_Y(y) &= f_{X_1}(x_1) \bigg|_{x_1=e^y} \cdot J \\ &= \frac{1}{\sqrt{2\pi} \sigma x_1} \exp \left\{ -\frac{1}{2\sigma^2} (\log x_1 - \mu)^2 \right\} \bigg|_{x_1=e^y} \cdot e^y \\ &= \frac{1}{\sqrt{2\pi} \sigma} e^{-y - \frac{1}{2\sigma^2} (y - \mu)^2 + y} \\ &= \frac{1}{\sqrt{2\pi} \sigma} \exp \left\{ -\frac{1}{2\sigma^2} (y - \mu)^2 \right\}; \quad y \in \mathbb{R} \end{aligned}$$

$\therefore Y \sim N(\mu, \sigma^2)$  distribution.

Now, let  $Y_1, \dots, Y_n$  are iid from  $N(\mu, \sigma^2)$  then

$$E(\bar{Y}_n) = E(Y_1) = \mu.$$

[To see (\*) observe that

$$\begin{aligned}
 E(Y_1) &= \mu + E(Y - \mu) = \mu + \int_{-\infty}^{\infty} \left( \frac{y - \mu}{\sigma} \right) \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left( \frac{y - \mu}{\sigma} \right)^2} dy \\
 &= \mu + \underbrace{\int_{-\infty}^{\infty} \frac{\sigma}{\sqrt{2\pi}} z e^{-z^2/2} dz}_0 \quad \text{where } z = \left( \frac{y - \mu}{\sigma} \right) \\
 &= \mu
 \end{aligned}
 ]$$

$$\therefore E\left(\frac{1}{n} \sum_{i=1}^n \log X_i\right) = \mu. \quad - (*)$$

Further,

$$E\left[\sum_{i=1}^n (Y_i - \bar{Y}_n)^2\right] = E\left[\sum_{i=1}^n Y_i^2 - n\bar{Y}_n^2\right] = (n-1)\sigma^2$$

[To see this, recall  $\frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \bar{Y})^2 \sim \chi_{(n-1)}^2$  — (I)]

$$\text{So, } E\left[\frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \bar{Y})^2\right] = (n-1)$$

$$\text{So, } E\left[\frac{1}{n-1} \sum_{i=1}^n (\log X_i)^2 - \frac{n}{n-1} \left\{ \frac{1}{n} \sum_{i=1}^n \log X_i \right\}^2\right] = \sigma^2.$$

— (\*\*)

From (\*) and (\*\*) we get

$\left\{ \frac{1}{n} \sum_{i=1}^n \log X_i, \frac{1}{n-1} \sum_{i=1}^n (\log X_i)^2 - \frac{1}{n(n-1)} \left( \sum_{i=1}^n \log X_i \right)^2 \right\}$  is the UMVUE of  $(\mu, \sigma^2)$  as they are unbiased estimator of  $(\mu, \sigma^2)$

and functions of CSS.

4. Compare the MLE and UMVUE estimators of  $\sigma^2$  for the log-normal distribution.

[5]

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n \left\{ \log X_i - \left( \frac{1}{n} \sum_{i=1}^n \log X_i \right) \right\}^2$$

$$\text{and } \hat{\sigma}_{UMVUE}^2 = \frac{1}{n-1} \sum_{i=1}^n \left\{ \log X_i - \left( \frac{1}{n} \sum_{i=1}^n \log X_i \right) \right\}^2$$

We will compare  $\hat{\sigma}_{MLE}^2$  and  $\hat{\sigma}_{UMVUE}^2$  by MSE.

$$\begin{aligned} \text{MSE}_{\sigma^2}(\hat{\sigma}_{MLE}^2) &= E \left[ (\hat{\sigma}_{MLE}^2 - \sigma^2)^2 \right] \\ &= \text{Bias}_{\sigma^2}^2(\hat{\sigma}_{MLE}^2) + \text{Var}(\hat{\sigma}_{MLE}^2). \end{aligned}$$

$$\begin{aligned} \text{Now, } E(\hat{\sigma}_{MLE}^2) &= E \left[ \frac{1}{n} \sum_{i=1}^n (\gamma_i - \bar{\gamma})^2 \mid \gamma \sim N(\mu, \sigma^2) \right] \\ &= \frac{1}{n} \cdot (n-1) \sigma^2 \quad [\text{by (I)}] \end{aligned}$$

$$\therefore \text{Bias}_{\sigma^2}^2(\hat{\sigma}_{MLE}^2) = \sigma^4 \left( \frac{n-1}{n} - 1 \right)^2 = \frac{\sigma^4}{n^2}$$

$$\text{Further, } \text{Var}(\hat{\sigma}_{MLE}^2) = \frac{1}{n^2} \text{Var} \left( \sum_{i=1}^n (\gamma_i - \bar{\gamma})^2 \right) = \frac{2\sigma^4(n-1)}{n^2}$$

[as if  $T \sim \chi_m^2$  then  $\text{Var}(T) = 2m$ ]

$$\therefore \text{MSE}_{\sigma^2}(\hat{\sigma}_{MLE}^2) = \sigma^4 \left[ \frac{1}{n^2} + \frac{2(n-1)}{n^2} \right] = \frac{2n-1}{n^2} \sigma^4 \quad \text{--- (II)}$$

$$\begin{aligned} \text{MSE}_{\sigma^2}(\hat{\sigma}_{UMVUE}^2) &= \text{Var}(\hat{\sigma}_{UMVUE}^2) \quad [\text{as unbiased}] \\ &= \frac{1}{(n-1)^2} \text{Var} \left( \sum_{i=1}^n (\gamma_i - \bar{\gamma})^2 \right) = \frac{2}{(n-1)} \sigma^4 \quad \text{--- (III)} \end{aligned}$$

Observe that, for any  $n \geq 1$ , we have  $\frac{2n-1}{n^2} < \frac{2}{n-1}$   
(as  $2n^2 - 3n + 1 < 2n^2 \Leftrightarrow n > 1/3$ ). So, from (I) and (II)  
 $MSE_{\sigma^2}(\hat{\sigma}_{MLE}^2) \leq MSE_{\sigma^2}(\hat{\sigma}_{UMVUE}^2)$ .

Hence,  $\hat{\sigma}_{MLE}^2$  is better estimator than  $\hat{\sigma}_{UMVUE}^2$ .

X — X

## Rough work